

Standard Deviation of a Standardized Vector

Devi Prasad M

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Assume $\mathbf{x}$ is not a constant vector, so $\text{std}(\mathbf{x}) \neq 0$. Therefore, the vector $\mathbf{z}$, the *standardized* version of $\mathbf{x}$, is well defined.

$$\begin{aligned}\mathbf{z} &= \frac{\tilde{\mathbf{x}}}{\text{std}(\mathbf{x})} \\ &= \frac{\sqrt{n}}{\|\tilde{\mathbf{x}}\|} \tilde{\mathbf{x}} \quad \because \text{std}(\mathbf{x}) = \frac{\|\tilde{\mathbf{x}}\|}{\sqrt{n}}\end{aligned}$$

Since $\mathbf{z}$ is a positive scalar multiple of $\tilde{\mathbf{x}}$, and scaling preserves the average,

$$\text{avg}(\mathbf{z}) = \frac{\sqrt{n}}{\|\tilde{\mathbf{x}}\|} \text{avg}(\tilde{\mathbf{x}}) = 0 \quad \because \text{avg}(\tilde{\mathbf{x}}) = 0$$

Let us try to determine $\text{std}(\mathbf{z})$.

$$\begin{aligned}\text{std}(\mathbf{z}) &= \frac{\|\tilde{\mathbf{z}}\|}{\sqrt{n}} \\ &= \frac{\|\mathbf{z}\|}{\sqrt{n}} \quad \because \text{avg}(\mathbf{z}) = 0 \\ &= \frac{1}{\sqrt{n}} \left\| \frac{\sqrt{n}}{\|\tilde{\mathbf{x}}\|} \tilde{\mathbf{x}} \right\| \\ &= \frac{1}{\sqrt{n}} \cdot \frac{\sqrt{n}}{\|\tilde{\mathbf{x}}\|} \cdot \|\tilde{\mathbf{x}}\| \quad \because \|c \mathbf{v}\| = |c| \|\mathbf{v}\| \text{ for } c > 0 \\ &= 1\end{aligned}$$